DAY-4

APTITUDE

Q) A = 10 days
B = 15 days
Together = ?

A's 1 day work = $\frac{1}{10}$
B's 1 day work = $\frac{1}{15}$

$A + B = \frac{1}{10} + \frac{1}{15}$.

LCM = 30.

$\frac{3}{30} + \frac{2}{30} \Rightarrow \frac{5}{30} \Rightarrow \frac{1}{6}$.

Together 6 days

Q) A = 6
B = 8
C = 12

A + B + C = ?

A 1's day work = $\frac{1}{6}$
B " = $\frac{1}{8}$
C " = $\frac{1}{12}$

$\therefore \frac{1}{6} + \frac{1}{8} + \frac{1}{12}$

$\Rightarrow \frac{4+3+9}{24}$

$\Rightarrow \frac{9}{24} \Rightarrow \frac{3}{8} \Rightarrow 2\frac{2}{3}$ days

Q) A alone 12 days

after 4 days B joins in 4 days

Time taken B = ?

A's 1 day work = $\frac{1}{12}$

work done in 4 day = $\frac{4}{12}$

$= \frac{1}{3}$

Remaining work = $1 - \frac{1}{3}$

$= \frac{2}{3}$

A+B ⇒ $\frac{2}{3}$ in 4 days

(A+B) 1 day = $\frac{2}{3} \div 4$

$= \frac{1}{6}$

$A = \frac{1}{12}$

$B = (A+B) - A$

$= \frac{1}{6} - \frac{1}{12}$

$= \frac{1}{12}$

B alone = 12 days

Q) work = workers × Day

5 × 8 = 40 units.

workers = x

x × 4 = 40

x = 10

Q) Filling pipe $= \frac{1}{6}$

Emptying pipe $= \frac{1}{8}$

Net work $= \frac{1}{6} - \frac{1}{8}$.

$$\frac{4-3}{24} \Rightarrow \frac{1}{24}$$

$\underline{T = 24 \text{ hours}}$